Pooja Desai

Materials Engineer

RESUME_1412

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Total Experience: 12 years

PROFESSIONAL SUMMARY

Materials Engineer with 12 years of experience in metallurgy, using Material Characterization, SEM, XRD, Metallography. Experienced in materials science, failure analysis and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Failure Analysis, SEM, Tensile Testing, DOE, ASTM Standards, XRD, Metallography, Quality Control, Material Characterization, Heat Treatment, Corrosion Testing

Methods: Materials science, Failure analysis, Quality assurance

PROFESSIONAL EXPERIENCE

Lead Materials Engineer | Meridian Vector Labs | Jan 2022 - Present

- Reduced material reject rate by 23% after heat-treatment window optimization
- Improved mechanical test repeatability by 26% through standard preparation
- Analyzed 365 metallurgical samples using documented acceptance criteria
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Materials Engineer | SparrowCloud Labs | Jan 2018 - Dec 2021

- Analyzed 261 metallurgical samples using documented acceptance criteria
- Extended accelerated corrosion resistance by 41% in comparative studies

Associate Materials Engineer | KoraLink Innovations | Jul 2014 - Dec 2017

- Reduced material reject rate by 26% after heat-treatment window optimization
- Improved mechanical test repeatability by 32% through standard preparation

PROJECTS

Alloy Heat-Treatment Study | Quality Control, DOE, Material Characterization, ASTM Standards

- Compared hardness, microstructure and treatment parameter stability.
- Documented acceptance criteria and evaluated the design using failure analysis and quality assurance.

Corrosion Resistance Trial | Quality Control, ASTM Standards, XRD, Material Characterization

- Tested coated coupons against accelerated exposure conditions.
- Documented acceptance criteria and evaluated the design using failure analysis and quality assurance.

EDUCATION

B.Tech in Metallurgical and Materials Engineering, Horizon Institute of Engineering, 2014

Relevant Methods: Materials science, Failure analysis, Quality assurance

CERTIFICATIONS

Materials Failure Analysis (2020)